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MAT-120

Quiz-2

Date: March 3, 2026

Name

Section

Instructions:

- Write your full name and section number.
- This quiz contains 2 problems. It is **YOUR responsibility** to check that your copy has all 2 problems.
- Make sure to read the instructions for each problem carefully.
- Use correct mathematical notation.
- Write cohesively and follow a logical sequence of steps.
- Check that your answer can be read and understood by the grader.

1. Consider the function

$$f(x) = \begin{cases} \dfrac{x^2 - 9}{x - 3}, & x \neq 3, \\ k, & x = 3. \end{cases}$$

   (a) Find the value of k that makes f continuous at $x = 3$.

   (b) State the type of discontinuity at $x = 3$ if $k \neq \lim_{x \to 3} f(x)$. Justify your answer.

   (c) Let $g(x) = x^3 - 4x + 1$. Use the Intermediate Value Theorem to show that $g(x) = 0$ has at least one real solution in the interval $[0, 1]$. Clearly verify all required conditions.

2. Let

$$h(x) = \frac{(x^3 - 2x)(x^2 + 1)}{x - 1}.$$

   (a) Find $h'(x)$ using the appropriate differentiation rules. Simplify your final answer.

   (b) Determine whether h is differentiable at $x = 1$. Justify your answer.

   (c) Let

$$p(x) = \begin{cases} x^2 + 2x, & x < 0, \\ 3x - 1, & x \geq 0. \end{cases}$$

   Determine whether $p(x)$ is differentiable at $x = 0$. Show all necessary steps.